

Ising model 1D

Paul Haskins

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Discrete model of spins ± 1 localised at specific points on a lattice.

$$H = - \sum_{ij}^N J_{ij} s_i s_j - \vec{\mu} \sum_i^N H_i s_i + \sum_{ijk}^N K_{ijk} s_i s_j s_k + \dots \quad (1)$$

$$s_i = \pm 1 \quad (2)$$

The first term is a new 'pairwise' interaction We call these all terms coupling constants.

$$J_{ij} = \begin{cases} > 0 \forall i, j & \text{if ferramagnetic} \\ < 0 \forall i, j & \text{anti-ferromagnetic} \\ g(x) & \text{otherwise} \\ 0, \text{non-interacting} \end{cases} \quad (3)$$

For the thermodynamic limit to exist for this system this condition must hold for at lease the pairwise coupling $\sum_{ij} |J_{ij}| < \infty$

We will typically study in canonical ensemble. For thermodynamic limit $N \gg 1$. What does it mean for the limit to exist? May not necessarily exist, previously we took the leading term but here when we take a limit it may not exist. We may have to impose a constraint like the one above.

We will make the simplification for now where we only study the nearest neighbor interactions. $J_{ij} = J > 0$.

$$H = -J \sum_{i=1} s_i s_{i+1} - \vec{\mu} \sum_{i=1} H_i s_i \quad (4)$$

Canonical Partition function (for diff B.C.)

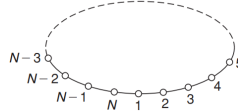


FIGURE 13.3 An Ising chain with a closed, endless structure.

Figure 1: 5.1.png

For higher Dimensions we can think of surface issues as B.C.

Now we will compute the free boundary condition H_i (an infinite chain).

$$\mathcal{Z} = \sum_{\{s_i\}} e^{\beta J \sum_{i=1}^{N-1} s_i s_{i+1}} \quad (5)$$

Note only up to $N - 1$ as the $i + 1$ term would account for the final spin.

$$\mathcal{Z} = \sum_{s_1} \dots \sum_{s_n} e^{\beta J (s_1 s_2 + s_2 s_3 + \dots + s_{N-1} s_N)} \quad (6)$$

Call $\eta_i = s_i s_{i+1} = \begin{cases} +1 & \text{if } s_i = s_{i+1} \\ -1 & \text{if } s_i \neq s_{i+1} \end{cases}$ We replace the $N-1$ sums by sums over η_i and keep the sum over s_i .

$$\Rightarrow \mathcal{Z} = \sum_{s_1} \sum_{\eta_1} \dots \sum_{\eta_{N-1}} e^{\beta J (\eta_1 + \eta_2 + \dots + \eta_{N-1})} \quad (7)$$

$$= \sum_{s_1 = \pm 1} \left[\prod_{i=1}^{N-1} \left(\sum_{\eta_i = \pm 1} e^{\beta J \eta_i} \right) \right] \quad (8)$$

$$= \sum_{s_1 = \pm 1} \left(\sum_{\eta = \pm 1} e^{\beta J \eta} \right)^{N-1} \quad (9)$$

$$= \sum_{s_1 = \pm 1} (2 \cosh(\beta J))^{N-1} \quad (10)$$

$$\Rightarrow \mathcal{Z} = 2 (2 \cosh(\beta J))^{N-1} \quad (11)$$

$H_i = 0$ free B.C. $\mathcal{Z} = 2 (2 \cosh(\beta J))^{N-1}$ periodic B.C.

Periodic B.C. we want last particle to interact with the first one

$$\mathcal{Z} = \sum_{\{s_i\}} e^{\beta J \sum_{i=1}^{N-1} s_i s_{i+1} + \beta J s_N s_1} \quad (12)$$

Suppose we introduce $\eta_i = s_i s_{i+1}$ How do we express $s_1 s_N$ in terms of η_i ?
we know the square must always be 1. so we can introduce these 1's.

$$s_N s_1 = s_1 s_2 s_2 s_3 s_3 \dots s_{N-1} s_{N-1} s_N = \eta_1 \dots \eta_{N-1} \quad (13)$$

Let us employ this change of variables $\{s_i\} \rightarrow \{\eta_i, s_1\}$.

$$\mathcal{Z} = \sum_{s_1} \sum_{\eta_1} \dots \sum_{\eta_{N-1}} e^{\beta J \eta_1} \dots e^{\beta J \eta_{N-1}} e^{\beta J \eta_1 \eta_2 \dots \eta_{N-1}} \quad (14)$$

$$= \sum_{s_1} \sum_{\eta_1} \dots \sum_{\eta_{N-1}} e^{\beta J \eta_1} e^{\beta J \eta_2} \dots e^{\beta J \eta_{N-1}} \cdot \sum_{\alpha=0}^{\infty} \frac{(\beta J)^\alpha}{\alpha!} (\eta_1 \eta_2 \dots \eta_{N-1})^\alpha \quad (15)$$

Exchanging the order of summation

$$\mathcal{Z} = \sum_{s_1} \sum_{\alpha=0}^{\infty} \frac{(\beta J)^\alpha}{\alpha!} \left(\sum_{\eta = \pm 1} e^{\beta J \eta} \eta^\alpha \right)^{N-1} \quad (16)$$

$$= \sum_{s_1 = \pm 1} \sum_{\alpha=0}^{\infty} \frac{(\beta J)^\alpha}{\alpha!} \left(e^{\beta J} + (-1)^\alpha e^{-\beta J} \right)^{N-1} \quad (17)$$

Note that

$$e^{\beta J} + (-1)^\alpha e^{-\beta J} = \begin{cases} \alpha = \text{even} & \text{if } 2 \cosh(\beta J) \\ \alpha = \text{odd} & 2 \sinh(\beta J) \end{cases} \quad (18)$$

$$\mathcal{Z} = \sum_{s_1 = \pm 1} \left[\sum_{n=0}^{\infty} \frac{(\beta J)^{2n}}{(2n)!} (2 \cosh(\beta J))^{N-1} + \sum_{n=0}^{\infty} \frac{(\beta J)^{2n+1}}{(2n+1)!} (2 \sinh(\beta J))^{N-1} \right] \quad (19)$$

$$= \sum_{s_1=\pm 1} [2^{N-1} \cdot \cosh^N(\beta J) + 2^{N-1} \sinh^N(\beta J)] \quad (20)$$

$$\Rightarrow \mathcal{Z} = 2^N \cosh^N(\beta J) + 2^N \sinh^N(\beta J) \quad (21)$$

Thus we have for periodic B.C.

$$\mathcal{Z} = (2 \cosh(\beta J))^N + (2 \sinh(\beta J))^N \quad (22)$$

We have the Helmholtz

$$\mathcal{F}(g_i, \beta, V, N) \quad (23)$$

where g_i is the coupling of the system.

We want to confirm the log of z behaves the same in the thermodynamic limit.

(a)

Free B.C.

$$\mathcal{F} = -kT \ln Z = -kT [\ln 2 + (N-1) \ln(2 \cosh(\beta J))] \quad (24)$$

$$= -kTN \left[\frac{1}{N} \ln 2 + \frac{N-1}{N} \ln(2 \cosh(\beta J)) \right] \rightarrow \mathcal{F}_A \approx -kTN \ln(2 \cosh(\beta J)) \quad (25)$$

for $N \gg 1$

(b)

Periodic B.C.

$$\mathcal{F} = -kT \ln [2^N \cosh^N(\beta J) [1 + \tanh^N(\beta J)]] \quad (26)$$

$$= -kTN \ln(2 \cosh(\beta J)) - kT \ln [1 + \tanh^N(\beta J)] \quad (27)$$

$$\mathcal{F}_B \approx -kTN \ln(2 \cosh(\beta J)) \quad (28)$$

for $N \gg 1$.

Where we have used

$$\ln [1 + \tanh^N(\beta J)] \approx \dots \quad (29)$$

$$0 < |\tanh(\beta J)| < 1 \quad (30)$$

So if we are in the thermodynamic limit we don't have to stick to one type of boundary conditions, we can pick the one which makes it easier for computations.

(c)

$H_i = H \neq 0$ Chose to study with periodic B.C.

Method: transfer matrix.

$$\mathcal{Z} = \sum_{s_1} \dots \sum_{s_N} e^{\beta H \sum_{i=1}^N + \beta J \sum_{i=1}^N s_i s_{i+1}} \quad (31)$$

$$= \sum_{s_1} \dots \sum_{s_N} e^{\frac{1}{2}\beta H(s_1+s_2)+\beta J s_1 s_2} \dots e^{\frac{1}{2}\beta H(s_N+s_1)+\beta J s_N s_1} \quad (32)$$

We will identify each of these terms with a component of a matrix element.

$$e^{\frac{1}{2}\beta H(s_1+s_2)+\beta J s_1 s_2} \equiv \mathcal{T}_{(s_1 s_2)} : \begin{bmatrix} e^{\beta H + \beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{-\beta H + \beta J} \end{bmatrix} \quad (33)$$

Substituting

$$\mathcal{Z} = \sum_{s_1} \sum_{s_N} \mathcal{T}_{(s_1 s_2)} \mathcal{T}_{(s_2 s_3)} \cdots \mathcal{T}_{(s_{N-1} s_N)} \mathcal{T}_{(s_N s_1)} \quad (34)$$

If we consider all the sums for all the terms we get a product

$$= \sum_{s_1} \left(\mathcal{T}^N \right)_{(s_1 s_1)} \quad (35)$$

where the sum is just the trace

$$\implies \mathcal{Z} = \text{Tr} [\mathcal{T}^N] \quad (36)$$

$$\mathcal{T} = \begin{bmatrix} e^{\beta(H+J)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-H)} \end{bmatrix} \quad (37)$$

$\mathcal{T}$ is real and symmetric $\rightarrow$ diagonalisable by some $\mathcal{O}$ with $\mathcal{O} = \mathcal{O}^T$. Then if

$$\mathcal{T}_{\text{diagonal}} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \quad (38)$$

$$\implies \mathcal{Z} = \text{Tr} [\mathcal{T}^N] = \lambda_1^N + \lambda_2^N \quad (39)$$

There are two cases either $\lambda_1 = \lambda_2$ or $\lambda_1 \neq \lambda_2$. where second case actually never happens. then if $\lambda_1 > \lambda_2$

$$\implies \mathcal{Z} = \lambda_1^N \left[1 + \left(\frac{\lambda_2}{\lambda_1} \right)^N \right] \quad (40)$$

$$\ln \mathcal{Z} = N \ln \lambda_1 + \ln \left(1 + \left(\frac{\lambda_2}{\lambda_1} \right)^N \right) \quad (41)$$

where the second term is bounded.

In the thermodynamic limit only **the largest** λ_i matters.

$$\lambda_{1,2} = e^{\beta J} \left[\cosh(\beta H) \pm \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right] \quad (42)$$

$$\implies \mathcal{F} = -NJ - NkT \ln \left[\cosh(\beta J) + \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right] \quad (43)$$

The first term is the energy, the log with sqrt means there may be non analytic $\mathcal{F}$ which leads to the potential for phase transitions.

In general if in the thermodynamic limit the thermodynamic potential exists (in our case Helmholtz) $\mathcal{F}(g_i)$, $i = 1, \dots, D$.

Phase diagram with D-axes.

Possible non-analyticities can be points, lines, planes etc.

Phase boundaries correspond necessarily to codimension-1 non-analytic regions. Codimensions is the total dimension of the system minus the dimension of the object.

The only non-analyticities are with the derivative not with $\mathcal{F}$ itself.